

Natural language statement

Domain

GroundTruth

2.0. The result [the second fundamental theorem] contains as a special case Picard's theorem that a transcendental meromorphic function assumes infinitely often all values in the plane except at most two. Notation: For a function f meromorphic in $|z| < R_0$, $m(r,a)$, $N(r,a)$ and $T(r,f) = m(r,\infty) + N(r,\infty)$ are Nevanlinna's proximity, counting and characteristic functions; $n(t,a)$ counts the roots of $f(z)=a$ in $|z| \leq t$ with multiplicity and $\bar{n}bar(t,a)$ counts them without. Correspondingly $\bar{N}bar(r,a) = \int_0^r n(t,a) dt + \bar{n}bar(0,a) \log r$. Assuming $T(r,f) \rightarrow \infty$ as $r \rightarrow R_0$, one sets $\delta(a) = \liminf_{r \rightarrow R_0} \{m(r,a)/(T(r))\}$, $\Theta(a) = 1 - \limsup_{r \rightarrow R_0} \{\bar{N}bar(r,a)/(T(r))\}$ and $\theta(a) = \liminf_{r \rightarrow R_0} \{N(r,a) - \bar{N}bar(r,a)/(T(r))\}$. f is "admissible" in $|z| < R_0$ when $R_0 = +\infty$ and f is not constant, or $R_0 < +\infty$ and (2.8) holds; $S(r,f)$ denotes any quantity satisfying the conclusions of Theorem 2.2, so $S(r,f) = o(T(r,f))$.

Complex analysis

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theorem hayman_2_0_picard_theorem (f : C -> C) (hf : Meromorphic f)
  (htr : not exists p q : Polynomial C, forall z, q.eval z != 0 -> f z =
    p.eval z / q.eval z) :
  exists a b : C, {c : C | not {z : C | f z = c}.Infinite} subset {a, b}
:= by
sorry
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Theorem 2.4. Suppose that $f(z)$ is admissible in $|z| < R_0$. Then the set of values a for which $\Theta(a) > 0$ is countable, and we have, on summing over all such values a $\sum_a \{\delta(a) + \theta(a)\} \leq \sum_a \Theta(a) \leq 2$. Notation: For a function f meromorphic in $|z| < R_0$, $m(r,a)$, $N(r,a)$ and $T(r,f) = m(r,\infty) + N(r,\infty)$ are Nevanlinna's proximity, counting and characteristic functions; $n(t,a)$ counts the roots of $f(z)=a$ in $|z| \leq t$ with multiplicity and $\bar{n}bar(t,a)$ counts them without. Correspondingly $\bar{N}bar(r,a) = \int_0^r n(t,a) dt + \bar{n}bar(0,a) \log r$. Assuming $T(r,f) \rightarrow \infty$ as $r \rightarrow R_0$, one sets $\delta(a) = \liminf_{r \rightarrow R_0} \{m(r,a)/(T(r))\}$, $\Theta(a) = 1 - \limsup_{r \rightarrow R_0} \{\bar{N}bar(r,a)/(T(r))\}$ and $\theta(a) = \liminf_{r \rightarrow R_0} \{N(r,a) - \bar{N}bar(r,a)/(T(r))\}$. f is "admissible" in $|z| < R_0$ when $R_0 = +\infty$ and f is not constant, or $R_0 < +\infty$ and (2.8) holds; $S(r,f)$ denotes any quantity satisfying the conclusions of Theorem 2.2, so $S(r,f) = o(T(r,f))$.

Complex analysis

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noncomputable def deficiency (f : C -> C) (a : WithTop C) : R :=
  liminf (fun r => proximity f a r / characteristic f top r) atTop

noncomputable def ramificationIndex (f : C -> C) (a : C) : R :=
  liminf (fun r => (logCounting f (a : WithTop C) r - reducedLogCounting f
    a r) /
    characteristic f top r) atTop

noncomputable def nevanlinnaTheta (f : C -> C) (a : C) : R :=
  1 - limsup (fun r => reducedLogCounting f a r / characteristic f top r)
    atTop

theorem hayman_2_4_deficiency_relation (f : C -> C) (hf : Meromorphic f)
  (hadm : Tendsto (characteristic f top) atTop atTop) :
  {a : C | 0 < nevanlinnaTheta f a}.Countable /\
  forall s : Finset C, (sum a in s, (deficiency f (a : WithTop C) +
    ramificationIndex f a)) <=
    sum a in s, nevanlinnaTheta f a /\
    (sum a in s, nevanlinnaTheta f a) <= 2 := by
sorry
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Theorem 2.5. If $f(z)$ is meromorphic and admissible in $|z| < R_0$ and $a_1(z)$, $a_2(z)$, $a_3(z)$ are distinct meromorphic functions satisfying for $\nu = 1, 2$, and 3 $T(r, a_\nu(z)) = o(T(r,f))$, as $r \rightarrow R_0$, then $\{1 + o(1)\}T(r,f) \leq \sum_{\nu=1}^3 \bar{N}bar(r, (1)/(f - a_\nu)) + S(r,f)$, as $r \rightarrow R_0$, where $S(r,f)$ satisfies the conclusions of Theorem 2.2. Notation: For a function f meromorphic in $|z| < R_0$, $m(r,a)$, $N(r,a)$ and $T(r,f) = m(r,\infty) + N(r,\infty)$ are Nevanlinna's proximity, counting and characteristic functions; $n(t,a)$ counts the roots of $f(z)=a$ in $|z| \leq t$ with multiplicity and $\bar{n}bar(t,a)$ counts them without. Correspondingly $\bar{N}bar(r,a) = \int_0^r n(t,a) dt + \bar{n}bar(0,a) \log r$. Assuming $T(r,f) \rightarrow \infty$ as $r \rightarrow R_0$, one sets $\delta(a) = \liminf_{r \rightarrow R_0} \{m(r,a)/(T(r))\}$, $\Theta(a) = 1 - \limsup_{r \rightarrow R_0} \{\bar{N}bar(r,a)/(T(r))\}$ and $\theta(a) = \liminf_{r \rightarrow R_0} \{N(r,a) - \bar{N}bar(r,a)/(T(r))\}$. f is "admissible" in $|z| < R_0$ when $R_0 = +\infty$ and f is not constant, or $R_0 < +\infty$ and (2.8) holds; $S(r,f)$ denotes any quantity satisfying the conclusions of Theorem 2.2, so $S(r,f) = o(T(r,f))$.

Complex analysis

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noncomputable def reducedLogCounting (f : C -> C) (a : C) (r : R) : R :=
  (integral t in Ioc 0 r, ((distinctCount f a t : R) - distinctCount f a 0)
    / t) +
  (distinctCount f a 0 : R) * Real.log r

theorem hayman_2_5_deficient_small_functions (f : C -> C) (hf : Meromorphic
  f)
  (hadm : Tendsto (characteristic f top) atTop atTop)
  (a : Fin 3 -> C -> C) (ha : forall nu, Meromorphic (a nu))
  (hdistinct : forall nu mu, nu != mu -> a nu != a mu)
  (hsmall : forall nu, Tendsto (fun r => characteristic (a nu) top r /
    characteristic f top r) atTop (nhds 0)) :
  forall eps : R, 0 < eps -> forallf r in atTop,
    (1 - eps) * characteristic f top r <=
      sum nu : Fin 3, reducedLogCounting (fun z => f z - a nu z) 0 r +
      eps * characteristic f top r := by
sorry
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Theorem 2.6. Suppose that $f_1(z)$, $f_2(z)$ are meromorphic in the plane and let $E_j(a)$ be the set of points z such that $f_j(z) = a$ ($j = 1, 2$). Then if $E_1(a) = E_2(a)$ for five distinct values of a , $f_1(z) = f_2(z)$, or f_1, f_2 are both constant. Notation: For a function f meromorphic in $|z| < R_0$, $m(r,a)$, $N(r,a)$ and $T(r,f) = m(r,\infty) + N(r,\infty)$ are Nevanlinna's proximity, counting and characteristic functions; $n(t,a)$ counts the roots of $f(z)=a$ in $|z| \leq t$ with multiplicity and $\bar{n}bar(t,a)$ counts them without. Correspondingly $\bar{N}bar(r,a) = \int_0^r n(t,a) dt + \bar{n}bar(0,a) \log r$. Assuming $T(r,f) \rightarrow \infty$ as $r \rightarrow R_0$, one sets $\delta(a) = \liminf_{r \rightarrow R_0} \{m(r,a)/(T(r))\}$, $\Theta(a) = 1 - \limsup_{r \rightarrow R_0} \{\bar{N}bar(r,a)/(T(r))\}$ and $\theta(a) = \liminf_{r \rightarrow R_0} \{N(r,a) - \bar{N}bar(r,a)/(T(r))\}$. f is "admissible" in $|z| < R_0$ when $R_0 = +\infty$ and f is not constant, or $R_0 < +\infty$ and (2.8) holds; $S(r,f)$ denotes any quantity satisfying the conclusions of Theorem 2.2, so $S(r,f) = o(T(r,f))$.

Complex analysis

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theorem hayman_2_6_five_value_theorem (f1 f2 : C -> C)
  (h1 : Meromorphic f1) (h2 : Meromorphic f2)
  (a : Fin 5 -> C) (ha : Function.Injective a)
  (hE : forall nu, {z : C | f1 z = a nu} = {z : C | f2 z = a nu}) :
  f1 = f2 \/ ((exists c1 : C, forall z, f1 z = c1) /\ exists c2 : C,
    forall z, f2 z = c2) := by
sorry
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Natural language statement	Domain	GroundTruth
$R_0\{Nbar(r,a)/(T(r))$ and $\theta(a) = \liminf_{r \rightarrow R_0\{N(r,a)-Nbar(r,a)/(T(r))\}}$. f is *admissible* in $z < R_0$ when $R_0 = +\infty$ and f is not constant, or $R_0 < +\infty$ and (2.8) holds; $S(r,f)$ denotes any quantity satisfying the conclusions of Theorem 2.2, so $S(r,f) = o\{T(r,f)\}$.		
Theorem 2.7. If $f(z)$ is a transcendental integral function then $f(z)$ possesses infinitely many fix-points of exact order n, except for at most one value of n. Notation: Let $f(z)$ be an integral function. Set $f_1(z) = f(z)$ and inductively $f_{\nu+1}(z) = \{f_{\nu}(z)\}$ for $\nu \geq 1$. The solutions of the equation $f_{\nu}(z) = z$ are called *fix-points of $f(z)$ of order ν*. If ζ is a fix-point of $f(z)$ of order ν, but of no lower order, then ζ is called a fix-point of *exact order* ν. An *integral function* is an entire function, and it is *transcendental* when it is not a polynomial.	Complex analysis	<pre>def IsTranscendentalEntire (f : C -> C) : Prop := Differentiable C f /\ not exists p : Polynomial C, forall z, f z = p.eval z theorem hayman_2_7_fixpoints_of_entire_functions (f : C -> C) (hf : IsTranscendentalEntire f) (iter : N -> C -> C) (h1 : forall z, iter 1 z = f z) (hstep : forall n z, iter (n + 1) z = f (iter n z)) : {n : N 1 <= n /\ not {z : C iter n z = z /\ forall m, 1 <= m -> m < n -> iter m z != z}.Infinite}.Subsingleton := by sorry</pre>
Theorem 2.9. Suppose that $f(z)$, $g(z)$ are integral functions and that $\phi(z) = g\{f(z)\}$ has finite order. Then either $f(z)$ is a polynomial or $g(z)$ has zero order. Notation: An *integral function* is an entire function. Writing $M(r,f) = \max_{ z =r} f(z)$, the function f has *finite order* when $\log M(r,f) = O(r^k)$ for some k, and *zero order* when $\log M(r,f) = O(r^\epsilon)$ for every $\epsilon > 0$.	Complex analysis	<pre>def HasFiniteOrder (f : C -> C) : Prop := exists k : R, forallf r in atTop, forall z : C, z <= r -> f z <= Real.exp (r ^ k) def HasZeroOrder (f : C -> C) : Prop := forall eps : R, 0 < eps -> forallf r in atTop, forall z : C, z <= r -> f z <= Real.exp (r ^ eps) theorem hayman_2_9_polya_composition_order (f g : C -> C) (hf : Differentiable C f) (hg : Differentiable C g) (hcomp : HasFiniteOrder (g comp f)) : (exists p : Polynomial C, forall z, f z = p.eval z) \/ HasZeroOrder g := by sorry</pre>
Theorem 3.4. Suppose that $f(z)$ is a transcendental meromorphic function in the plane and $\psi(z) = f^*(l)\{z\}$. Then in the notation of Theorem 2.4 we have $\sum_{a \neq \infty} \Theta_a(a,\psi) \leq 1 + (1)/(l+1)$. In particular $\psi(z)$ assumes every finite value with at most one exception infinitely often. Notation: For a function f meromorphic in $z < R_0$, $m(r,a)$, $N(r,a)$ and $T(r,f) = m(r,\infty) + N(r,\infty)$ are Nevanlinna's proximity, counting and characteristic functions; $n(t,a)$ counts the roots of $f(z) = a$ in $z \leq t$ with multiplicity and $nbar(t,a)$ counts them without. Correspondingly $Nbar(r,a) = \int_0^r nbar(t,a) dt + nbar(0,a) \log r$. Assuming $T(r,f) \rightarrow \infty$ as $r \rightarrow R_0$, one sets $\delta(a) = \liminf_{r \rightarrow R_0} \{m(r,a)/(T(r))\}$, $\Theta(a) = 1 - \limsup_{r \rightarrow R_0} \{Nbar(r,a)/(T(r))\}$ and $\theta(a) = \liminf_{r \rightarrow R_0} \{N(r,a)-Nbar(r,a)/(T(r))\}$. f is *admissible* in $z < R_0$ when $R_0 = +\infty$ and f is not constant, or $R_0 < +\infty$ and (2.8) holds; $S(r,f)$ denotes any quantity satisfying the conclusions of Theorem 2.2, so $S(r,f) = o\{T(r,f)\}$.	Complex analysis	<pre>noncomputable def nevanlinnaTheta (f : C -> C) (a : C) : R := 1 - limsup (fun r => reducedLogCounting f a r / characteristic f top r) atTop theorem hayman_3_4_derivative_deficiency_bound (f : C -> C) (hf : Meromorphic f) (htr : not exists p q : Polynomial C, forall z, q.eval z != 0 -> f z = p.eval z / q.eval z) (l : N) (hl : 1 <= l) (psi : C -> C) (hpsi : psi = iteratedDeriv l f) : (forall s : Finset C, (sum a in s, nevanlinnaTheta psi a) <= 1 + 1 / (l + 1 : R)) /\ {a : C not {z : C psi z = a}.Infinite}.Subsingleton := by sorry</pre>
Corollary. For all sufficiently large l, $f^*(l)\{z\}$ has zeros in every disk in which $f(z)$ is meromorphic and has at least two distinct poles. Notation: For a function f meromorphic in $z < R_0$, $m(r,a)$, $N(r,a)$ and $T(r,f) = m(r,\infty) + N(r,\infty)$ are Nevanlinna's proximity, counting and characteristic functions; $n(t,a)$ counts the roots of $f(z) = a$ in $z \leq t$ with multiplicity and $nbar(t,a)$ counts them without. Correspondingly $Nbar(r,a) = \int_0^r nbar(t,a) dt + nbar(0,a) \log r$. Assuming $T(r,f) \rightarrow \infty$ as $r \rightarrow R_0$, one sets $\delta(a) = \liminf_{r \rightarrow R_0} \{m(r,a)/(T(r))\}$, $\Theta(a) = 1 - \limsup_{r \rightarrow R_0} \{Nbar(r,a)/(T(r))\}$ and $\theta(a) = \liminf_{r \rightarrow R_0} \{N(r,a)-Nbar(r,a)/(T(r))\}$. f is *admissible* in $z < R_0$ when $R_0 = +\infty$ and f is not constant, or $R_0 < +\infty$ and (2.8) holds; $S(r,f)$ denotes any quantity satisfying the conclusions of Theorem 2.2, so $S(r,f) = o\{T(r,f)\}$.	Complex analysis	<pre>theorem hayman_3_6_corollary_derivative_zeros_in_disk : forallf l in atTop, forall (f : C -> C) (z0 : C) (R : R), 0 < R -> (forall z in Metric.ball z0 R, MeromorphicAt f z) -> (exists p in Metric.ball z0 R, exists q in Metric.ball z0 R, p != q /\ not AnalyticAt C f p /\ not AnalyticAt C f q) -> exists z in Metric.ball z0 R, iteratedDeriv l f z = 0 := by sorry</pre>

Natural language statement

Domain

GroundTruth

Theorem 3.6. Suppose that $f(z)$ is meromorphic in $|z - z_0| < R$, where $0 < R \leq \infty$, and has at least two distinct poles there. Let r be the radius of the largest circle with centre z_0 containing no pole of $f(z)$ in its interior. Then (i) if the circle $|z - z_0| = r$ contains at least two distinct poles of $f(z)$, then for every positive δ , the equation $f^{(l)}(z) = 0$ has roots in $|z - z_0| < \delta$, when l is sufficiently large; (ii) if the circle $|z - z_0| = r$ contains only one pole of $f(z)$, then if δ is sufficiently small, $f^{(l)}(z) \rightarrow \infty$ as $l \rightarrow \infty$ uniformly in $|z - z_0| \leq \delta$. Notation: For a function f meromorphic in $|z| < R_0$, $m(r, a)$, $N(r, a)$ and $T(r, f) = m(r, \infty) + N(r, \infty)$ are Nevanlinna's proximity, counting and characteristic functions; $n(t, a)$ counts the roots of $f(z) = a$ in $|z| \leq t$ with multiplicity and $\bar{n}(t, a)$ counts them without. Correspondingly $\bar{N}(r, a) = \int_0^r n(t, a) dt + \bar{n}(0, a) \log r$. Assuming $T(r, f) \rightarrow \infty$ as $r \rightarrow R_0$, one sets $\delta(r) = \liminf_{r \rightarrow R_0} \{m(r, a) / T(r)\}$, $\Theta(r) = 1 - \limsup_{r \rightarrow R_0} \{\bar{N}(r, a) / T(r)\}$ and $\theta(r) = \liminf_{r \rightarrow R_0} \{N(r, a) - \bar{N}(r, a) / T(r)\}$. f is "admissible" in $|z| < R_0$ when $R_0 = \infty$ and f is not constant, or $R_0 < \infty$ and (2.8) holds; $S(r, f)$ denotes any quantity satisfying the conclusions of Theorem 2.2, so $S(r, f) = o\{T(r, f)\}$.

Complex analysis

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theorem hayman_3_6_derivative_zeros_near_poles (f : C -> C) (z0 : C) (R : R) (hR : 0 < R)
(hf : forall z in Metric.ball z0 R, MeromorphicAt f z)
(P : Set C) (hP : P = {z in Metric.ball z0 R | not AnalyticAt C f z})
(htwo : exists p in P, exists q in P, p != q) (r : R) (hr : 0 < r)
(hrmax : IsGreatest {t : R | 0 < t /\ forall z in P, not ||z - z0|| < t} r) :
(exists p in P, exists q in P, p != q /\ ||p - z0|| = r /\ ||q - z0|| = r) ->
  forall delta : R, 0 < delta -> forall l in atTop,
    exists z in Metric.ball z0 delta, iteratedDeriv l f z = 0) /\
  ((exists! p, p in P /\ ||p - z0|| = r) -> forall delta in nhds[>] (0 : R),
    Tendsto (fun l => z : Metric.closedBall z0 delta, ||iteratedDeriv l f z||) atTop atTop) := by
sorry
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Theorem 3.8. Suppose that $f(z)$ is meromorphic and has only a finite number of poles in the plane, and that $f(z)$, $f^{(l)}(z)$ have only a finite number of zeros for some $l \geq 2$. Then $f(z) = (P_1(z))/(P_2(z))e^{A(z)}$, where P_1, P_2, P_3 are polynomials. If, further, $f(z)$ and $f^{(l)}(z)$ have no zeros, then $f(z) = e^{A(z)+B}$ or $f(z) = (Az+B)^{-n}$. Notation: For a function f meromorphic in $|z| < R_0$, $m(r, a)$, $N(r, a)$ and $T(r, f) = m(r, \infty) + N(r, \infty)$ are Nevanlinna's proximity, counting and characteristic functions; $n(t, a)$ counts the roots of $f(z) = a$ in $|z| \leq t$ with multiplicity and $\bar{n}(t, a)$ counts them without. Correspondingly $\bar{N}(r, a) = \int_0^r n(t, a) dt + \bar{n}(0, a) \log r$. Assuming $T(r, f) \rightarrow \infty$ as $r \rightarrow R_0$, one sets $\delta(r) = \liminf_{r \rightarrow R_0} \{m(r, a) / T(r)\}$, $\Theta(r) = 1 - \limsup_{r \rightarrow R_0} \{\bar{N}(r, a) / T(r)\}$ and $\theta(r) = \liminf_{r \rightarrow R_0} \{N(r, a) - \bar{N}(r, a) / T(r)\}$. f is "admissible" in $|z| < R_0$ when $R_0 = \infty$ and f is not constant, or $R_0 < \infty$ and (2.8) holds; $S(r, f)$ denotes any quantity satisfying the conclusions of Theorem 2.2, so $S(r, f) = o\{T(r, f)\}$.

Complex analysis

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theorem hayman_3_8_tumura_clunie_form (f : C -> C) (hf : Meromorphic f) (l : N) (hl : 2 <= l)
(hpoles : {z : C | not AnalyticAt C f z}.Finite)
(hzeros : {z : C | f z = 0}.Finite)
(hlzeros : {z : C | iteratedDeriv l f z = 0}.Finite) :
(exists P1 P2 P3 : Polynomial C, P2 != 0 /\
  forall z, P2.eval z != 0 -> f z = P1.eval z / P2.eval z *
  Complex.exp (P3.eval z)) /\
  (({z : C | f z = 0} = empty /\ {z : C | iteratedDeriv l f z = 0} = empty) ->
  (exists A B : C, forall z, f z = Complex.exp (A * z + B)) /\
  exists (A B : C) (n : N), 1 <= n /\ forall z, A * z + B != 0 -> f z = (A * z + B) ^ (-(n : Z))) := by
sorry
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